

Long-Term Surgical Outcomes of Temporal Lobe Epilepsy Associated With Low-Grade Brain Tumors

Ji Hoon Phi, MD¹; Seung-Ki Kim, MD, PhD¹; Byung-Kyu Cho, MD, PhD¹; Seo Young Lee, MD²; Su Yeon Park, MD¹; Sung-joon Park, MD¹; Sang Kun Lee, MD, PhD³; Ki Joong Kim, MD, PhD⁴; and Chun Kee Chung, MD, PhD¹

BACKGROUND: Tumor-related temporal lobe epilepsy (TLE) has a high likelihood of medical intractability and requires surgical treatment. The aims of this study were to analyze the long-term surgical outcomes of and to present appropriate surgical strategies for tumor-related TLE. **METHODS:** The clinical data of 87 consecutive patients diagnosed with tumor-related TLE were analyzed. The median age at surgery was 22 years. Sixteen patients had a tumor confined to the amygdala or the parahippocampal gyrus, and 10 of them received a tailored lesionectomy without hippocampectomy. The surgical outcome was evaluated based on 3 aspects: seizure control, tumor control, and discontinuation of antiepileptic drugs (AEDs). **RESULTS:** The actuarial seizure and tumor control rates at the fifth year postoperatively were 79% and 90%, respectively. Seizure control was highly correlated with tumor control. The following factors were found to be significantly associated with poor seizure control: duration of epilepsy >10 years, presence of a remote focus on surface electroencephalography, and incomplete tumor removal. The actuarial AED maintenance rates were 47% at the second year and 11% at the fifth year. The median time to AED discontinuation was 22 months. A younger age at surgery was found to be significantly associated with an increased chance of AED discontinuation. Tailored resection focusing on the tumor resulted in a favorable outcome, even for tumors confined to the amygdala or the parahippocampal gyrus. **CONCLUSIONS:** Surgical treatment of tumor-related TLE resulted in long-term seizure control in the majority of patients. Maximal tumor removal can be recommended for tumor-related TLE. **Cancer** 2009;115:5771-9. © 2009 American Cancer Society.

KEY WORDS: brain tumor, temporal lobe epilepsy, tumor control, seizure control, tailored resection.

Temporal lobe epilepsy (TLE) is the prototype of the surgically remediable epilepsy syndromes. Approximately 10% to 30% of TLE cases are associated with a brain tumor, the majority of which are low-grade glial tumors.^{1,2} Tumor-related TLE has a high likelihood of medical intractability, and early surgical treatment is recommended.³

Corresponding author: Chun Kee Chung, MD, PhD, Department of Neurosurgery, Seoul National University College of Medicine, 101 Daehangno, Jongno-gu, Seoul, 110-744, Republic of Korea; Fax: (011) 82-2-744-8459; chungc@snu.ac.kr

¹Department of Neurosurgery, Seoul National University College of Medicine, Seoul, Republic of Korea; ²Department of Neurology, Kangwon National University College of Medicine, Chuncheon, Republic of Korea; ³Department of Neurology, Seoul National University College of Medicine, Seoul, Republic of Korea; ⁴Department of Pediatrics, Seoul National University College of Medicine, Seoul, Republic of Korea

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The surgical outcomes for tumor-related TLE are comparable to those for TLE with mesial temporal sclerosis (MTS); adequate seizure control can be achieved in >80% of the patients.^{4,5} However, the long-term control of seizures and of the tumor is a concern in these patients, and to the best of our knowledge, very few longitudinal analyses have been reported in the literature. Furthermore, data regarding the discontinuation of antiepileptic drugs (AEDs) after surgery for tumor-related TLE are scarce.

The extent of surgery necessary in tumor-related epilepsy has been controversial. In tumor-related TLE, the tumor's anatomic proximity to and even involvement of the mesial temporal lobe, which is the most epileptogenic structure in the human brain, render it more difficult to determine the optimal surgical extent.^{6,7} In the current study, we retrospectively reviewed the clinical outcomes of 87 patients with tumor-related TLE. We estimated the actuarial seizure control rates and analyzed the relevant prognostic variables related to seizure control and AED discontinuation in these patients. In addition, we investigated the effects of the anatomic location of tumors and of the extent of surgery on the seizure outcomes.

MATERIALS AND METHODS

Patients

We selected patients who had surgery at the epilepsy surgery program of the Seoul National University Hospital from 1995 to 2006, based on a diagnosis of focal epilepsy with a primary brain tumor in the temporal lobe. A minimum postoperative follow-up of 1 year was required. We excluded patients with an extra-axial tumor or a malignant tumor that corresponded to World Health Organization grade 3 or 4. Eighty-seven patients were included in this study. The clinical and radiologic data of the patients were retrospectively reviewed. This study was approved by the institutional review board of the Seoul National University Hospital.

Forty-nine patients were male, and 38 were female. The median age at surgery was 22 years (range, 1-62 years). The median age at seizure onset was 14 years (range, 4 months-62 years). The mean duration of epilepsy was 66 months (range, 1 month-48 years). The patients received a comprehensive presurgical evaluation

that consisted of magnetic resonance imaging (MRI), interictal surface electroencephalography (EEG), ictal video-EEG monitoring, positron emission tomography, single-photon emission tomography, and neuropsychologic tests. Fifty-one (59%) patients received the Wada test for determination of language and memory lateralization.

Seizure Characteristics and EEG Findings

Seventy-six (87%) patients had complex partial seizures, and 13 (15%) patients demonstrated simple partial seizures. Secondary generalization was documented in 55 (63%) patients. The mean seizure frequency was 32 seizures/month (range, 0.2-600 seizures/month). The mean number of AEDs taken by the patients at the time of the surgery was 1.7 (range, 1-4 AEDs).

Interictal surface EEG was available for review for 60 patients. Interictal EEG was localized to the affected temporal lobe in 29 (48%) patients, to the contralateral temporal lobe in 1 (2%) patient, to the bilateral temporal lobes in 3 (5%) patients, and to the affected temporal lobe and an extratemporal lobe in 4 (7%) patients. It was nonlocalizable in 23 (38%) patients. Ictal video-EEG monitoring was available for review for 51 patients. Ictal EEG was localized to the affected temporal lobe in 36 (71%) patients, to the bilateral temporal lobes in 2 (4%) patients, and to the affected temporal lobe and other foci in 9 (18%) patients. Ictal EEG was nonlocalizable in 4 (8%) patients.

Invasive EEG was obtained for 7 patients using grids and strips. Six patients had a tumor in the lateral temporal lobe (4 were anterolateral and 2 were posterolateral), and 1 had a tumor in the mesial temporal lobe. The surgical extent was eventually extended in 6 patients according to the results of the invasive EEG analysis.

Tumors

Fifty-seven (66%) patients had a brain tumor in the left temporal lobe, and 30 (34%) patients had a tumor in the right temporal lobe. Tumor location was divided into the mesial, anterolateral, and posterior temporal lobes (Fig. 1). Forty-nine (56%) tumors were located in the mesial temporal lobe, 25 (29%) tumors were located in the anterolateral temporal lobe, and 13 (15%) tumors were

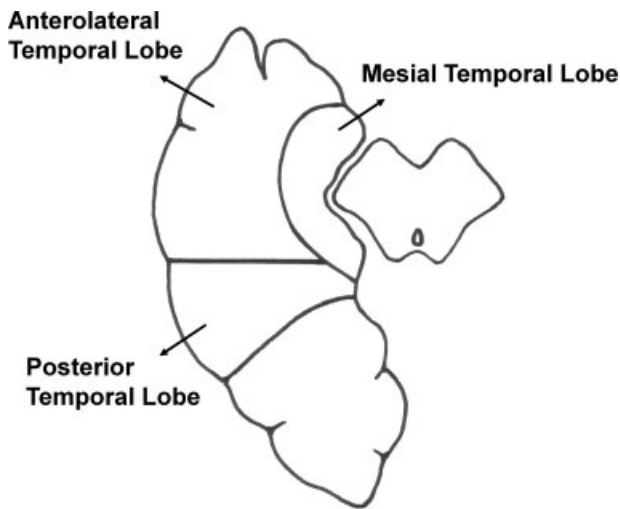


FIGURE 1. Locations of tumors in patients with temporal lobe epilepsy are shown. The location of the tumors was divided into the mesial, anterolateral, and posterior temporal lobes. The boundary between the anterolateral and posterior temporal lobes was set at an arbitrary line across the posterior margin of the midbrain on axial magnetic resonance imaging.

found in the posterior temporal lobe. Dysembryoplastic neuroepithelial tumor (DNET) and ganglioglioma were the most common tumors, and were diagnosed in 29 (33%) and 25 (28%) patients, respectively. The pathologic diagnoses of the tumors are summarized in Table 1.

Dual Pathology

Concomitant atrophy and high signal intensity in the hippocampus in fluid attenuated inversion recovery images (dual pathology) were detected on preoperative MRI in 8 patients. The abnormal hippocampus was located on the same side as the tumor in all patients. Five patients had a tumor in the mesial temporal lobe, mainly in the parahippocampal gyrus, and 3 had a tumor in the anterolateral temporal lobe. One patient with a tumor in the dominant anterolateral temporal lobe underwent invasive monitoring, and ictal EEG was localized to the ipsilateral mesial temporal lobe. Invasive monitoring was not performed for the other 2 patients with a tumor in the nondominant anterolateral temporal lobe.

In all 8 patients, both the tumor and the abnormal hippocampus were removed simultaneously, regardless of the tumor extent. Hippocampal sclerosis was confirmed on pathological examination in 7 patients.

Table 1. Pathologic Diagnosis of Tumors From 87 Patients

Pathologic Diagnosis	No. of Cases
Dysembryoplastic neuroepithelial tumor	29
Ganglioglioma	25
Oligodendroglioma	11
Astrocytoma	8
Pleomorphic xanthoastrocytoma	7
Ependymoma	2
Gangliocytoma	1
Desmoplastic infantile ganglioglioma	1
Choroid plexus papilloma	1
Intracortical sclerosing meningioma	1
Atypical glioneuronal tumor	1
Total	87

Dual pathology was missed preoperatively in 1 patient. After tailored resection without removal of the affected hippocampus, the patient received a reoperation.

Extent of Surgery

The extent of surgery was determined based on several factors: tumor location and structures involved, laterality of the lesion (dominant or nondominant hemisphere), imaging evidence of dual pathology, and electrophysiologic evidence of other epileptogenic zones (Fig. 2).

In 16 patients, the tumor was confined to the amygdala or to the parahippocampal gyrus (in 11 patients and 5 patients, respectively), without apparent involvement of the hippocampus. In a dominant hemisphere, the hippocampus was preserved, unless tailored lesionectomy was technically difficult. In a nondominant hemisphere, the entire mesial temporal lobe was removed.

Outcome Evaluation and Statistical Analyses

In all patients, the extent of surgery and degree of tumor removal were evaluated using MRI results recorded within 3 months postoperatively. Macroscopic total resection was achieved in 74 (85%) patients. No adjuvant treatment was immediately given to the patients with an incompletely resected tumor, except for 1 patient with a large residual oligodendroglioma who received postoperative radiotherapy. AEDs were prescribed to the patients as preoperatively and were tapered down after the patients became seizure free. The mean follow-up period was

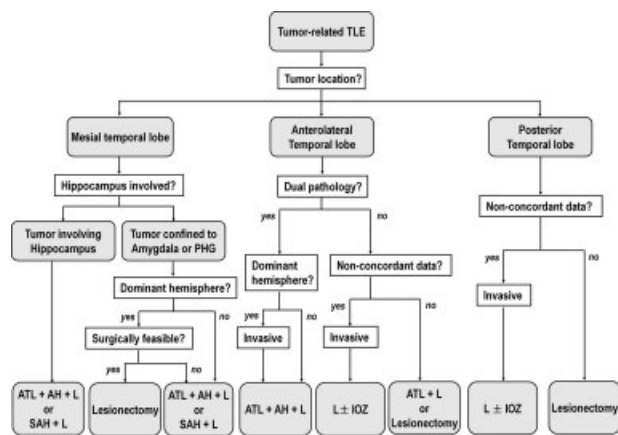


FIGURE 2. Treatment algorithm for tumor-related temporal lobe epilepsy (TLE) as applied to the current study population is shown. PHG indicates parahippocampal gyrus; ATL, anterior temporal lobectomy; AH, amygdalohippocampectomy; L, lesionectomy; SAH, selective amygdalohippocampectomy; IOZ, ictal onset zone.

54 months (range, 12–128 months). Three treatment outcomes were measured: seizure control, tumor control, and AED discontinuation. The status of seizure control was evaluated by chart review and was classified according to Engel's classification.⁸ The factors affecting seizure control and AED discontinuation were analyzed using the Kaplan-Meier method and the Cox proportional hazards model. The significance level applied was 5%. Patient-related factors (sex, age at seizure onset, duration of epilepsy, age at surgery, secondary generalization, multiple ictal foci on EEG, and dual pathology on MRI), tumor-related factors (laterality, mesial/anterolateral/posterior location, and tumor involvement), and treatment-related factors (degree of tumor resection and extent of surgery) were included in the analyses.

RESULTS

Seizure Outcome

In the Kaplan-Meier survival analysis of overall seizure control, the endpoint was defined as the time when seizure recurred after surgery. The actuarial seizure control rates were 92% at the first year, 86% at the second year, and 79% at the fifth year postoperatively (Fig. 3A).

In univariate analyses of dichotomous variables using the Kaplan-Meier model, a duration of epilepsy of

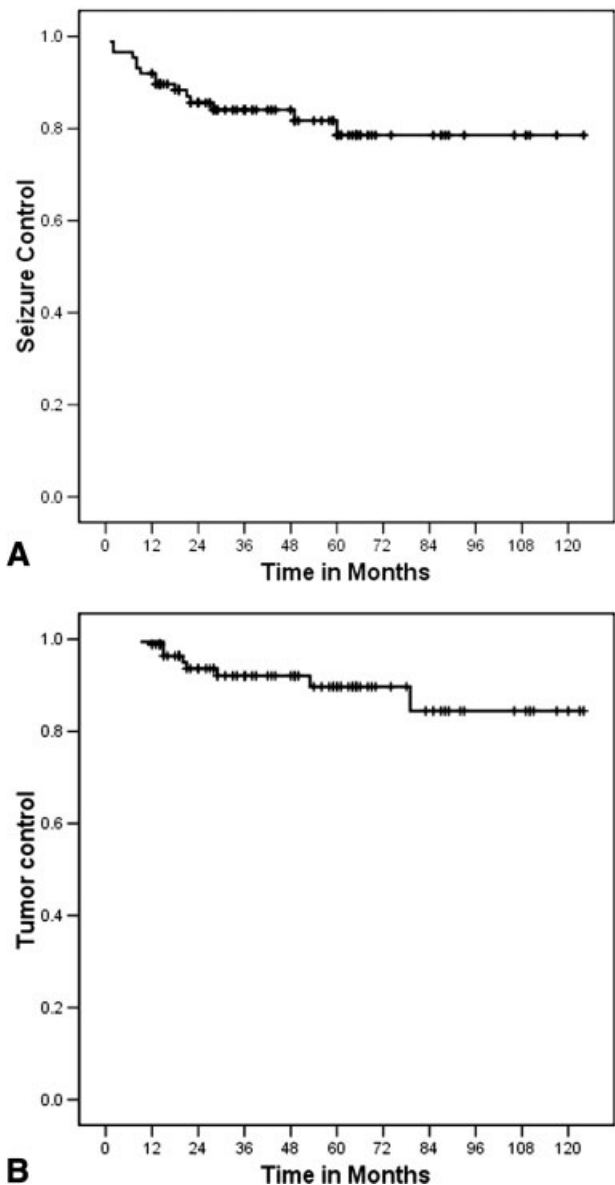


FIGURE 3. Kaplan-Meier plots illustrate the crude cumulative incidences of (A) seizure control and (B) tumor control. The actuarial seizure control rates were 92%, 86%, and 79%, respectively, at the first, second, and fifth years. Seizure control and tumor control were found to be highly correlated (odds ratio, 10.1; $P = .002$ [Fisher exact test]).

≥ 10 years, the presence of a remote focus on surface EEG, and failure of total tumor removal were related to seizure control failure ($P = .0001$, $P = .001$, and $P = .014$, respectively; log-rank test). Tumor location in the mesial temporal lobe was associated with improved seizure outcome when compared with tumors located in the

Table 2. Relative Risks for Seizure Control Failure Estimated Using a Cox Proportional Hazards Model

Factors	Univariate Analysis			Multivariate Analysis		
	<i>P</i>	Crude RR	95% CI	<i>P</i>	Adjusted RR	95% CI
Age at surgery	.375	1.015	0.982-1.050	.774	0.993	0.947-1.041
Duration of epilepsy ≥ 10 y	<.001	6.865	2.484-18.975	<.001	11.648	2.970-45.681
Presence of a remote EEG focus	.003	4.920	1.737-13.938	.018	5.499	1.338-12.606
Failure of macroscopic total resection	.022	3.655	1.206-11.082	<.001	18.220	3.812-87.090
Mesial temporal tumor location	.046	0.334	0.114-0.979	.133	0.361	0.095-1.364

RR indicates relative risk; 95% CI, 95% confidence interval; EEG, electroencephalography.

Table 3. Relative Risks for AED Discontinuation Estimated Using a Cox Proportional Hazards Model in 72 Patients Without Seizure Recurrence

Factors	Univariate Analysis			Multivariate Analysis		
	<i>P</i>	Crude RR	95% CI	<i>P</i>	Adjusted RR	95% CI
Age at surgery	.002	0.963	0.941-0.987	.003	0.965	0.942-0.988
Duration of epilepsy ≥ 10 y	.858	0.917	0.364-2.314	NA	NA	NA
Presence of a remote EEG focus	.379	0.631	0.227-1.759	NA	NA	NA
Failure of macroscopic total resection	.074	0.346	0.108-1.108	.112	0.388	0.121-1.247
Mesial temporal tumor location	.741	1.093	0.645-1.851	NA	NA	NA

AED indicates antiepileptic drug; RR, relative risk; 95% CI, 95% confidence interval; NA, not applicable; EEG, electroencephalography.

anterolateral and posterior temporal lobes ($P = .035$; log-rank test).

In univariate analyses using a Cox proportional hazards model, the same 4 factors were found to be significantly associated with seizure outcomes (Table 2). Multivariate analyses performed using the same model demonstrated that a duration of epilepsy of ≥ 10 years, the presence of a remote focus on surface EEG, and failure of total tumor removal were significantly associated with poor seizure outcomes, and that tumor location in the mesial temporal lobe was marginally correlated with improved seizure outcome (Table 2).

Tumor Outcome

For analysis of overall tumor control, the endpoint was defined as the time when recurrence of a totally removed tumor or progression of a partially resected tumor were noticed on follow-up MRI. The actuarial tumor control rates were 99% at the first year, 94% at the second year, and 90% at the fifth year postoperatively (Fig. 3B). Seizure control and tumor control were found to be highly correlated ($P = .002$; odds ratio, 10.1; Fisher exact test).

Tumor recurrence was observed in 8 patients. The initial pathologic diagnoses were DNET, pleomorphic xanthoastrocytoma, ganglioglioma, and oligodendroglioma (2 cases for each). Although DNETs are known to have a low recurrence potential, 2 of 4 patients with incomplete removal of a DNET experienced disease recurrence after 20 months and 29 months, respectively. Failure of total tumor removal was the single significant predictive factor of tumor recurrence ($P < .0001$; log-rank test).

AED Discontinuation

Kaplan-Meier survival analyses revealed that the actuarial AED maintenance rates in 72 patients without seizure recurrence were 79% at the first year, 47% at the second year, and 11% at the fifth year. The median time to AED discontinuation was 22 months. Univariate and multivariate analyses using a Cox proportional hazards model revealed that a younger age at surgery was significantly associated with a greater chance of AED discontinuation (Table 3).

Table 4. Seizure Outcomes According to the Tumor Location, Involvement, and Type of Surgery

Tumor Location	Tumor Involvement	No. of Patients	Type of Surgery	Seizure Control		No. of Patients
				Yes	No	
Mesial temporal	A	11	ATL+A+L	3	3	6
			A+L	2	0	2
			ATL+AH+L	3	0	3
	A+H	9	SAH+L	1	0	1
			ATL+AH+L	8	0	8
	A+H+PHG	12	ATL+AH+L	8	1	9
			SAH+L	3	0	3
	A+H+PHG+latT	4	ATL+A+L	1	0	1
			ATL+AH+L	2	1	3
	H+PHG	6	ATL+AH+L	6	0	6
	PHG	5	ATL+AH+L	2	0	2
			SAH+L	1	0	1
			ATL+L	2	0	2
ATL+AH+L			2	0	2	
PHG+FG			2	ATL+AH+L	2	0
Anterolateral temporal	Ant-latT	25	ATL+AH+L	5	2	7
			ATL+L	7	1	8
			L	8	1	9
			L+IOZ	0	1	1
Posterior temporal	Post-medT	4	ATL+AH+L	0	1	1
			L	1	2	3
	Post-latT	9	ATL+A+L	1	0	1
			L	4	1	5
Total			L+IOZ	2	1	3
				72	15	87

A indicates amygdala (amygdalectomy); ATL, anterior temporal lobectomy; L, lesionectomy; AH, amygdalohippocampectomy; H, hippocampus (hippocampectomy); SAH, selective amygdalohippocampectomy; PHG, parahippocampal gyrus; latT, lateral temporal; FG, fusiform gyrus; Ant, anterior; IOZ, ictal onset zone; Post, posterior; medT, medial temporal.

Dual Pathology

The 8 patients who received mesial resection based on the preoperative MRI findings indicative of dual pathology were seizure-free postoperatively, with the exception of 1 patient who developed seizure control failure after 13 months without tumor recurrence. The patient responded to a change in AEDs and finally became Engel Class 2.

Surgical Strategy

Seizure control rates according to the location of tumors and surgical extent are summarized in Table 4. The surgical outcomes in patients with a posteromedial temporal lobe tumor appeared to be poor (seizure control failure in 3 of 4 patients), partly because the tumors were incompletely removed in these 3 patients.

Surgical resection was tailored to the extent of tumor involvement in 49 patients with a tumor in the mesial temporal lobe. The hippocampus was always removed in

cases in which the tumor involved this structure, with the exception of 1 patient with a ganglioglioma who refused hippocampectomy for fear of memory deterioration. The tumor recurred with malignant transformation, and the patient died of the disease.

The hippocampus was preserved in 8 patients with a tumor confined to the amygdala. Five patients became seizure free. Two patients underwent reoperation (mainly for hippocampectomy) after invasive monitoring and subsequently became seizure free. A hippocampus was also preserved in 2 patients with a small tumor involving only the parahippocampal gyrus, and seizures were controlled in these patients.

Dual pathology was overlooked in a patient with a ganglioglioma confined to the left amygdala; thus, tailored tumor resection alone was performed initially. The patient underwent reoperation (hippocampectomy) after seizure recurrence. Hippocampal sclerosis was pathologically confirmed, and the patient became seizure free after the reoperation (Fig. 4).

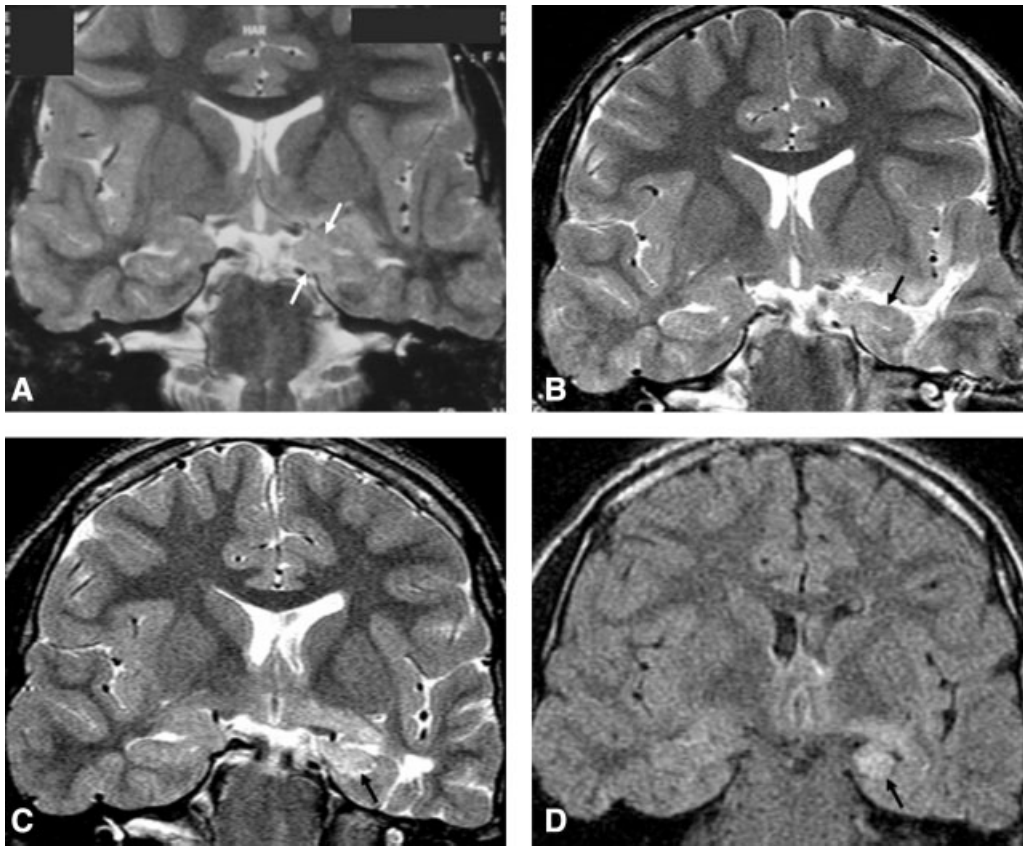


FIGURE 4. (A) A 13-year-old boy who had intractable epilepsy for 2 years was found to have a ganglioglioma (arrows) in the left amygdala detected on magnetic resonance imaging (MRI). (B) A tailored resection (temporopolar resection plus amygdalectomy including the tumor) was performed. Intraoperatively, the hippocampus appeared free from tumor invasion and was left untouched. Immediate postoperative MRI revealed a slightly smaller hippocampus (arrow) in the left side without a definite signal abnormality in T2-weighted images. Seizures recurred 21 months postoperatively, and MRI demonstrated a clearly atrophic hippocampus (arrow) with high signal intensity in (C) T2-weighted and (D) fluid attenuated inversion recovery images. Invasive monitoring revealed ictal discharges from the left hippocampus. The patient underwent hippocampectomy, and hippocampal sclerosis was found on pathologic examination.

DISCUSSION

Seizure and Tumor Outcome

Tumor-related TLE appears to have a favorable prognosis. Seizure control rates after surgery, including both Engel Class 1 and 2 outcomes, have been reported to be approximately 80%^{4,5}; however, previous studies on tumor-related TLE provided seizure control rates at a fixed time point. In the current study, we calculated the actuarial rates for seizure control, because epilepsy per se as well as brain tumors are dynamic disease entities that require a long-term follow-up. The actuarial seizure control rates were 86% at the second year and 79% at the fifth year postoperatively, and there was no seizure recurrence after

the fifth year. The stable maintenance of seizure outcome in tumor-related epilepsy patients has been reported in another study.⁴ This stability is in contrast with the steady decline in seizure control rates after surgery for MTS.^{9,10}

Three clinical factors were found to be significantly associated with seizure outcome in the present study. Although some authors reported that good seizure control can be obtained despite incomplete tumor resection,^{11,12} the importance of total tumor removal for seizure control has been emphasized by many authors, especially in recent studies.^{4,5,13} However, Daumas-Duport et al¹⁴ reported that they observed no recurrence in 17 patients with an incompletely resected DNET; in contrast, we found 2 patients with a recurring DNET who had seizure

recurrence. Moreover, a patient died of relentless recurrence and malignant transformation of low-grade ganglioglioma after intended subtotal resection for protection of memory function. Although it has been proposed that brain tumors presenting with chronic epilepsy are biologically indolent,^{4,15} to the best of our knowledge, the prognosis cannot yet be predicted on a patient basis, and total tumor removal has an important meaning in the oncological perspective.

In the current study, a longer duration of epilepsy was correlated with poor seizure outcomes, which is consistent with previous studies.^{4,5,13} Tumor-related TLE may attain treatment resistance if it persists for a long time. Because a remote focus on EEG was also associated with a poor seizure outcome in this and other studies,^{4,16} the 2 phenomena—long duration of epilepsy and generation of a secondary epileptogenic focus—may be interrelated.

AED Discontinuation

To our knowledge, few studies to date regarding tumor-related epilepsy have addressed AED use in the postoperative period. In a study on tumor-related TLE, 17 of 44 seizure-free patients were completely off AEDs.⁵ Only 4 of 26 children with tumor-related TLE were able to quit AEDs, although they attained a seizure-free state.¹⁷ In the current study, the actuarial rate of AED discontinuation at the fifth year in seizure-controlled patients was 89%. The higher value found in this study may reflect differences in the statistical analysis. Younger patients appeared to be more prone to AED discontinuation. Young patient age at surgery is also known to be a favorable prognostic factor of AED withdrawal after surgery for MTS.¹⁸

Surgical Strategy

The surgical extent necessary and sufficient for seizure control in tumor-related epilepsy remains highly controversial.^{12,17,19} In tumor-related TLE, the surgical procedure may be limited to lesionectomy (tumor resection) or be extended up to the standard temporal lobectomy. To the best of our knowledge, little evidence is available regarding this issue, mainly because well-matched comparisons of surgical procedures are difficult to undertake in these heterogeneous patients. It was reported that the prognosis of tumor-related epilepsy was determined not

by the surgery type, but by the tumor pathology.⁴ A detailed surgical strategy has been proposed: a lesionectomy and/or anterior temporal lobectomy (ATL) for lateral temporal lobe tumors, and a lesionectomy plus ATL and/or amygdalohippocampectomy for mesial temporal lobe tumors. The performance of amygdalohippocampectomy was limited to a nondominant hemisphere or to dual pathologic entities in a dominant hemisphere.¹⁶ Our strategy was similar to this, with the difference that we applied a more tailored lesionectomy to mesial temporal lobe tumors. Approximately 70% of our patients who received a tailored lesionectomy for an amygdala or a parahippocampal gyrus tumor had a good seizure outcome; however, 2 patients required removal of the hippocampus after tailored lesionectomy, and MTS was found in 1 of the patients. Therefore, thorough evaluation of neuroimaging and electrophysiologic results should be undertaken before tailored lesionectomy, especially in the case of amygdala tumors.

In the case of anterolateral temporal lobe tumors, 75% of patients in the current study received lesionectomy and/or ATL, and 25% underwent ATL plus amygdalohippocampectomy. The outcome was slightly worse in the latter group. The latter group may represent a more complex disease, with the presence of dual pathologic lesions. Similar results were reported in a study on TLE patients; only 1 patient had poor seizure outcome of 27 patients with a lateral temporal lobe tumor who received lesionectomy (13 patients) or ATL (14 patients), whereas 2 of 7 patients who received lateral tumor resection with amygdalohippocampectomy had poor outcomes.²⁰

Tumors in the posterior temporal lobe can be regarded as extratemporal tumors in many aspects. In the current study, seizure outcome was mainly affected by the completeness of tumor removal; therefore, a lesionectomy could be a safe and effective procedure for these tumors.

Limitations and Advantages of This Study

Many efforts to establish clinical standards in the treatment of tumor-related epilepsy have been discouraged by the small number of patients, long inclusion periods, the heterogeneity of the disease, and uncontrolled study designs. On the basis of retrospective analyses, this study had similar limitations. Although we focused on tumor-related

TLE patients, the study population was heterogeneous in age distribution, tumor pathology, and the extent of the tumor. The treatment strategy was not concretely uniform but heavily individualized. Moreover, the lack of comparative neuropsychologic analyses lessens the significance of tailored resection with preservation of the hippocampus, which was applied in a small group of patients. Nonetheless, this study presents substantial advantages. This is a single-institutional study in which the patients were treated in a relatively short period, were all examined using modern neuroimaging technologies, and were operated on by 3 surgeons. At the time of last follow-up, 60 (69%) patients were being observed in outpatient clinics, and long-term evaluation of the outcome was possible. Furthermore, longitudinal analyses were conducted to overcome the shortcomings of the cross-sectional approaches that have frequently been applied to this subject.

Conflict of Interest Disclosures

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